

# Rigorous Bounds for Covering the Unit Disk by 100 Equal Disks

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## Abstract

Let  $r_D(N)$  be the smallest radius for which  $N$  equal closed disks can cover the unit disk. We give an explicit construction of 100 equal disks whose union contains a disk of radius  $\sqrt{67}$ , proving

$$r_D(100) \leq \frac{1}{\sqrt{67}} = 0.122169444356305223\dots$$

The construction is based on a triangular lattice. Its covering property is reduced to a finite certificate of elementary lattice triangles; the certificate and the verification code are included in the repository. We also specialize the asymptotic theorem of Birgin, Gardenghi, and Laurain to the unit disk and collect rigorous lower and constructive upper bounds for every  $N \leq 100$ . The exact global value of  $r_D(100)$  remains open in the published literature.

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# 1 Introduction

The disk covering problem asks for the smallest common radius for which  $N$  congruent disks can cover the unit disk. In the notation of this paper,

$$r_D(N) = \inf_{c_1, \dots, c_N \in \mathbb{R}^2} \max_{x \in \mathbb{D}} \min_{1 \leq i \leq N} \|x - c_i\|_2, \quad \mathbb{D} = \{x \in \mathbb{R}^2 : \|x\|_2 \leq 1\}. \quad (1)$$

The exact values are known only for small  $N$ ; the cases  $N \leq 10$  are classical, while the global optimality question is open in general for larger  $N$  [4, 2, 3].

The purpose of this paper is twofold. First, we prove a verified explicit upper bound for  $N = 100$ . Second, we record a general asymptotic law and rigorous finite- $N$  bounds, so that the discussion is not based on a black-box numerical simulation.

The main construction uses the triangular lattice generated by

$$u = (\sqrt{3}, 0), \quad v = \left( \frac{\sqrt{3}}{2}, \frac{3}{2} \right). \quad (2)$$

For an integer pair  $(i, j)$ , write

$$q(i, j) = i^2 + ij + j^2, \quad c_{i,j} = iu + jv. \quad (3)$$

We use the 97 lattice points

$$A = \{(i, j) \in \mathbb{Z}^2 : q(i, j) \leq 27\} \quad (4)$$

and add three arbitrary lattice points of the next shell to obtain 100 centers. The central result is

$$\mathbb{D}_{\sqrt{67}} \subseteq \bigcup_{(i,j) \in A} \overline{B}(c_{i,j}, 1). \quad (5)$$

Thus, after scaling by  $1/\sqrt{67}$ , the unit disk is covered by 100 disks of radius  $1/\sqrt{67}$ .

The proof of (5) consists of two independent pieces:

- (i) a finite Delaunay-triangle certificate, showing that every point of  $\mathbb{D}_{\sqrt{67}}$  lies in an elementary triangle having at least one vertex in  $A$ ;
- (ii) a finite critical-ray certificate, showing that the construction cannot cover a disk of radius larger than  $\sqrt{67}$ .

The certificates are generated by the reproducible scripts in `scripts/`.

The paper is organized as follows. Section 2 establishes the elementary scaling and area bounds. Section 3 proves the  $N = 100$  upper bound. Section 4 states the general asymptotic theorem and specializes it to the unit disk. Section 5 describes the complete  $N \leq 100$  table. Section 6 discusses the distinction between the constructed upper bound and a global optimum.

## 2 Preliminaries

Let  $D_R = \{x \in \mathbb{R}^2 : \|x\|_2 \leq R\}$ . We write  $\overline{B}(c, r)$  for the closed disk of radius  $r$  centered at  $c$ .

**Lemma 2.1** (Scaling). *Suppose  $N$  unit disks cover  $D_R$ . Then  $N$  disks of radius  $1/R$  cover  $\mathbb{D}$ . Consequently,*

$$r_D(N) \leq \frac{1}{R}. \quad (6)$$

*Proof.* Let the unit disks have centers  $c_1, \dots, c_N$  and suppose  $D_R \subseteq \bigcup_i \overline{B}(c_i, 1)$ . Apply the linear map  $x \mapsto x/R$ . It sends  $D_R$  onto  $\mathbb{D}$  and  $\overline{B}(c_i, 1)$  onto  $\overline{B}(c_i/R, 1/R)$ . Hence

$$\mathbb{D} \subseteq \bigcup_i \overline{B}(c_i/R, 1/R).$$

This proves the claim.  $\square$

**Lemma 2.2** (Area lower bound). *For every  $N \geq 1$ ,*

$$r_D(N) \geq \frac{1}{\sqrt{N}}. \tag{7}$$

*Proof.* If  $N$  disks of radius  $r$  cover  $\mathbb{D}$ , then their union has area at least that of  $\mathbb{D}$ . Their total area is at most  $N\pi r^2$ , whence  $N\pi r^2 \geq \pi$  and  $r \geq 1/\sqrt{N}$ .  $\square$

**Remark 2.3.** For  $N = 100$ , Lemma 2.2 gives  $r_D(100) \geq 1/10$ . The construction below gives  $r_D(100) \leq 1/\sqrt{67}$ ; the exact value is not claimed.

### 3 The explicit construction for $N = 100$

#### 3.1 Lattice shells

Recall the triangular lattice  $L = \{iu + jv : i, j \in \mathbb{Z}\}$  generated by (2). The quadratic form  $q(i, j) = i^2 + ij + j^2$  is the squared norm of  $iu + jv$  up to the factor 3:

$$\|iu + jv\|_2^2 = 3q(i, j). \tag{8}$$

The number of lattice points in each shell is as follows.

Table 1: Shells of the triangular lattice used in (4).

| $q$ | number of lattice points | cumulative count |
|-----|--------------------------|------------------|
| 0   | 1                        | 1                |
| 1   | 6                        | 7                |
| 3   | 6                        | 13               |
| 4   | 6                        | 19               |
| 7   | 12                       | 31               |
| 9   | 6                        | 37               |
| 12  | 6                        | 43               |
| 13  | 12                       | 55               |
| 16  | 6                        | 61               |
| 19  | 12                       | 73               |
| 21  | 12                       | 85               |
| 25  | 6                        | 91               |
| 27  | 6                        | 97               |

**Lemma 3.1.** *The set  $A$  defined in (4) has exactly 97 elements.*

*Proof.* This follows from the shell count in Table 1. For completeness, the shell counts can be verified directly from the quadratic form: the numbers of representations of the listed values of  $q$  are exactly the entries in the second column.  $\square$

Let  $T$  be any three lattice points in the next shell, for example

$$T = \{(-6, 2), (4, -6), (2, 4)\}. \quad (9)$$

Then  $|A \cup T| = 100$ . The three extra points play no role in the covering proof; they are included only to make the number of centers exactly 100.

### 3.2 Delaunay triangles

The triangular lattice has a Delaunay triangulation by equilateral triangles of side  $\sqrt{3}$ . Every such triangle has circumradius 1. Therefore, if a point  $x$  lies in a lattice triangle one of whose vertices is  $c \in A$ , then  $\|x - c\|_2 \leq 1$ .

We use the following finite certificate.

**Definition 3.2.** An *elementary lattice triangle* is a triangle whose three vertices are lattice points and whose three sides have length  $\sqrt{3}$ .

**Lemma 3.3** (Finite triangle certificate). *Every elementary lattice triangle that meets  $D_{\sqrt{67}}$  has at least one vertex in  $A$ .*

*Proof.* The proof is a finite enumeration, recorded in `data/delaunay_triangles.csv`. If an elementary triangle meets  $D_{\sqrt{67}}$ , its diameter is  $\sqrt{3}$ , so each vertex is at distance at most  $\sqrt{67} + \sqrt{3} < \sqrt{67} + 2$  from the origin. Thus only lattice points in the finite disk of radius  $\sqrt{67} + 2$  can occur.

The verification script enumerates all elementary triangles with vertices in this finite set. It finds 204 triangles meeting the target disk. Among them,

- 156 have all three vertices in  $A$ ;
- 36 have exactly two vertices in  $A$ ;
- 12 have exactly one vertex in  $A$ ;
- none have all three vertices outside  $A$ .

The twelve boundary cases are listed in Table 2. Consequently every such triangle has a vertex in  $A$ .  $\square$

Table 2: The twelve elementary triangles with exactly one vertex in  $A$ . The middle entry in each row is the selected  $A$ -vertex.

| triangle vertices              | selected $A$ -vertex | triangle vertices           |
|--------------------------------|----------------------|-----------------------------|
| $(-6, 1), (-6, 2), (-5, 1)$    | $(-5, 1)$            | $(5, -4), (6, -5), (6, -4)$ |
| $(-6, 4), (-6, 5), (-5, 4)$    | $(-5, 4)$            | $(4, 1), (4, 2), (5, 1)$    |
| $(-5, -1), (-4, -2), (-4, -1)$ | $(-4, -1)$           | $(1, -6), (1, -5), (2, -6)$ |
| $(-5, 6), (-4, 5), (-4, 6)$    | $(-4, 5)$            | $(1, 4), (1, 5), (2, 4)$    |
| $(-2, -4), (-1, -5), (-1, -4)$ | $(-1, -4)$           | $(4, -6), (4, -5), (5, -6)$ |
| $(-2, 6), (-1, 5), (-1, 6)$    | $(-1, 5)$            | $(5, -1), (6, -2), (6, -1)$ |

**Theorem 3.4** (Coverage). *The 97 unit disks centered at the lattice points of  $A$  cover the disk of radius  $\sqrt{67}$ :*

$$\mathbb{D}_{\sqrt{67}} \subseteq \bigcup_{(i,j) \in A} \overline{B}(c_{i,j}, 1). \quad (10)$$

*Proof.* Let  $x \in \mathbb{D}_{\sqrt{67}}$ . Since the elementary lattice triangles tile the plane,  $x$  lies in some elementary triangle  $T$ . This triangle meets  $\mathbb{D}_{\sqrt{67}}$  and hence, by Lemma 3.3, has a vertex  $c_{i,j}$  with  $(i, j) \in A$ . The circumradius of an elementary triangle is 1, so  $\|x - c_{i,j}\|_2 \leq 1$ . Therefore  $x$  belongs to the corresponding unit disk. This proves (10).  $\square$

### 3.3 Sharpness of the constructed radius

Coverage alone gives only  $R(100) \geq \sqrt{67}$ . We also record that the construction does not cover a disk of any larger radius. This is a property of the fixed construction, not a global optimality assertion.

**Lemma 3.5** (Critical ray). *For the construction  $A$ , the largest centered disk contained in the union of the unit disks is exactly  $\mathbb{D}_{\sqrt{67}}$ .*

*Proof.* Set

$$p = \left( -\frac{7\sqrt{3}}{2}, -\frac{11}{2} \right), \quad e = \frac{p}{\|p\|_2}, \quad \|p\|_2 = \sqrt{67}. \quad (11)$$

For a center  $c$ , the intersection of the ray  $\{te : t \geq 0\}$  with  $\overline{B}(c, 1)$  is the interval

$$I_c = \left[ c \cdot e - \sqrt{1 - \|c - (c \cdot e)e\|_2^2}, c \cdot e + \sqrt{1 - \|c - (c \cdot e)e\|_2^2} \right], \quad (12)$$

whenever  $\|c - (c \cdot e)e\|_2 \leq 1$ , and is empty otherwise. The finite list of intervals for the centers in  $A$  is recorded in `data/critical_ray_intervals.csv`. Their union contains  $[0, \sqrt{67}]$ , but no interval on this ray extends beyond  $\sqrt{67}$ . Indeed the maximal right endpoint is attained by the centers

$$c_{2,3} = \left( \frac{7\sqrt{3}}{2}, \frac{9}{2} \right), \quad c_{1,4} = (3\sqrt{3}, 6), \quad (13)$$

and equals  $\sqrt{67}$  exactly. Hence a centered disk of radius larger than  $\sqrt{67}$  cannot be contained in the union.  $\square$

**Corollary 3.6.** *We have*

$$r_D(100) \leq \frac{1}{\sqrt{67}} = 0.122169444356305223 \dots \quad (14)$$

*Proof.* By Theorem 3.4, 97 unit disks already cover  $\mathbb{D}_{\sqrt{67}}$ . Adding the three remaining centers can only increase the union. Lemma 2.1 with  $R = \sqrt{67}$  gives (14).  $\square$

## 4 The general asymptotic law

The finite construction above is deliberately elementary. For the general behavior as  $N$  grows, we use the following theorem of Birgin, Gardenghi, and Laurain [1].

**Theorem 4.1** (BGL asymptotic bounds). *Let  $A \subset \mathbb{R}^2$  be a bounded set satisfying the polygonal-arc regularity class of [1], let  $\text{Vol}(A)$  be its area, let  $\text{Per}(\partial A)$  be its perimeter, and let  $\bar{k}$  be its number of boundary arcs. There exists  $\bar{m} \in \mathbb{N}$  with*

$$\bar{m} \geq \frac{16\pi\bar{k}}{3\sqrt{3}}$$

*such that, for every  $m > \bar{m}$ , the optimal radius  $r_A^*(m)$  for covering  $A$  by  $m$  minimum-radius identical disks satisfies*

$$r_A^*(m) = \sqrt{\frac{2\text{Vol}(A)}{3\sqrt{3}m}} + R(m), \quad \underline{R}(m) \leq R(m) \leq \bar{R}(m), \quad (15)$$

where

$$\underline{R}(m) = -\frac{2\text{Per}(\partial A)}{3\sqrt{3}m} - \frac{8\pi\bar{k}(2\text{Vol}(A))^{1/2}}{(3\sqrt{3}m)^{3/2}}, \quad (16)$$

$$\bar{R}(m) = \frac{2\mu_m\text{Per}(\partial A) + 8\pi\bar{k}\mu_m^2}{(2\text{Vol}(A)3\sqrt{3}m)^{1/2}}, \quad (17)$$

$$\mu_m = \frac{2\text{Per}(\partial A) + [4\text{Per}(\partial A)^2 + 2\text{Vol}(A)(3\sqrt{3}m - 10\pi\bar{k})]^{1/2}}{3\sqrt{3}m - 16\pi\bar{k}}. \quad (18)$$

In particular,

$$r_A^*(m) = \sqrt{\frac{2\text{Vol}(A)}{3\sqrt{3}m}} + O\left(\frac{1}{m}\right). \quad (19)$$

The theorem is stated here for reference; its proof is contained in [1]. We now specialize it to the unit disk.

**Corollary 4.2** (Disk specialization). *For the unit disk,*

$$\text{Vol}(\mathbb{D}) = \pi, \quad \text{Per}(\partial\mathbb{D}) = 2\pi, \quad \bar{k} = 1. \quad (20)$$

Put

$$T_N = 3\sqrt{3}N. \quad (21)$$

Then, for every  $N > \bar{m}$ ,

$$\underline{r}(N) \leq r_D(N) \leq \bar{r}(N), \quad (22)$$

where

$$\underline{r}(N) = \sqrt{\frac{2\pi}{T_N}} - \frac{4\pi}{T_N} - \frac{8\pi\sqrt{2\pi}}{T_N^{3/2}}, \quad (23)$$

$$\bar{r}(N) = \sqrt{\frac{2\pi}{T_N}} + \frac{4\pi\mu_N + 8\pi\mu_N^2}{\sqrt{2\pi T_N}}, \quad (24)$$

$$\mu_N = \frac{4\pi + \sqrt{16\pi^2 + 2\pi(T_N - 10\pi)}}{T_N - 16\pi}. \quad (25)$$

In particular,

$$r_D(N) = \sqrt{\frac{2\pi}{3\sqrt{3}N}} + O\left(\frac{1}{N}\right). \quad (26)$$

*Proof.* Substitute (20) into (16)–(18). The leading term becomes

$$\sqrt{\frac{2\pi}{3\sqrt{3}N}},$$

and both remainders are  $O(1/N)$ . This proves the stated bounds and (26).  $\square$

## 5 Bounds for every $N \leq 100$

For  $N \leq 10$ , the exact values are classical and are recorded in Table 3. For  $N \geq 11$ , we use

$$\max \left\{ \frac{1}{\sqrt{N}}, r(N) \right\} \leq r_D(N) \leq (\text{explicit construction}). \quad (27)$$

The lower bound is the maximum of the area bound (Lemma 2.2) and the BGL lower bound (Corollary 4.2). The upper bound is supplied by the constructions:

- the known constructions tabulated by Friedman [3] for  $11 \leq N \leq 25$ ;
- the triangular-lattice shell construction described in Section 3 for  $26 \leq N \leq 100$ .

The complete machine-readable table is `data/r_D_N1-100_bounds.csv`. Selected values are shown in Table 4.

Table 3: Exact values for  $N \leq 10$ .

| $N$ | value                    | decimal              |
|-----|--------------------------|----------------------|
| 1   | 1                        | 1                    |
| 2   | 1                        | 1                    |
| 3   | $\sqrt{3}/2$             | 0.866025403784438647 |
| 4   | $\sqrt{2}/2$             | 0.707106781186547524 |
| 5   | algebraic root           | 0.609382864080709655 |
| 6   | algebraic root           | 0.555905211416588705 |
| 7   | $1/2$                    | 0.5                  |
| 8   | $1/(1 + 2 \cos(2\pi/7))$ | 0.445041867912628809 |
| 9   | $\sqrt{2} - 1$           | 0.414213562373095049 |
| 10  | $1/(1 + 2 \cos(2\pi/9))$ | 0.394930843634698458 |

## 6 Discussion and status

The construction in Section 3 proves

$$r_D(100) \leq \frac{1}{\sqrt{67}}.$$

The area bound of Lemma 2.2 gives

$$r_D(100) \geq \frac{1}{10}.$$

Table 4: Selected rigorous intervals from the complete  $N \leq 100$  table.

| $N$ | lower bound    | upper bound    | status               |
|-----|----------------|----------------|----------------------|
| 25  | 0.2            | 0.239146311738 | known construction   |
| 30  | 0.182574185835 | 0.277350098113 | lattice construction |
| 40  | 0.158113883008 | 0.2            | lattice construction |
| 50  | 0.141421356237 | 0.188982236505 | lattice construction |
| 60  | 0.129099444874 | 0.164356535502 | lattice construction |
| 70  | 0.119522860933 | 0.152498570333 | lattice construction |
| 80  | 0.111803398875 | 0.142857142857 | lattice construction |
| 90  | 0.105409255339 | 0.128020416345 | lattice construction |
| 100 | 0.1            | 0.122169444356 | lattice construction |

Thus every claimed exact value must lie in

$$[0.1, 0.122169444356305223 \dots].$$

The two endpoints do not coincide. The exact global value of  $r_D(100)$  is not known in the published literature, and this paper does not claim it.

The general asymptotic theorem explains the shape of the table: the leading term is the hexagonal-lattice term, and the first correction is of order  $1/N$ . The lattice construction used here is a finite realization of that asymptotic geometry. For small  $N$ , other configurations, notably ring configurations, can perform better; this is why the table does not apply one formula mechanically to all indices.

All numerical data in the paper are reproducible. In particular, the  $N = 100$  coverage proof uses a finite triangle certificate rather than a black-box global optimization or a GPU search.

## A Shell count

Formula (8) shows that the shell of index  $q$  is the set of integer pairs representing  $q$  by the positive definite quadratic form  $i^2 + ij + j^2$ . The entries in Table 1 are the standard representation counts for the listed values. The cumulative count through  $q = 27$  is

$$1 + 6 + 6 + 6 + 12 + 6 + 6 + 12 + 6 + 12 + 12 + 6 + 6 = 97.$$

## B Finite triangle certificate

The script `scripts/verify_certificate.py` performs the following finite check:

1. generate all lattice points within distance  $\sqrt{67} + 2$  of the origin;
2. enumerate all elementary triangles with pairwise distance  $\sqrt{3}$ ;
3. retain the triangles whose convex hull meets  $D_{\sqrt{67}}$ ;
4. check that each retained triangle has at least one vertex in  $A$ .

The resulting detailed list is stored in `data/delaunay_triangles.csv`. The certificate reports 204 triangles and no triangle with all three vertices outside  $A$ .

## C Critical-ray certificate

The file `data/critical_ray_intervals.csv` contains all nonempty intervals (12) for the direction (11). The maximal endpoint is the exact value  $\sqrt{67}$ . The equality is attained by the centers (13).

## D Reproducibility

From the repository root, run

```
python scripts/verify_certificate.py
python scripts/generate_bounds.py
powershell -ExecutionPolicy Bypass -File scripts/build-paper.ps1
```

The scripts require only Python 3 and standard scientific libraries already listed in the repository. No GPU computation is used.

## References

- [1] Ernesto G. Birgin, John L. Gardenghi, and Antoine Laurain. Asymptotic bounds on the optimal radius when covering a set with minimum radius identical balls. *Mathematics of Operations Research*, 49(3):1855–1889, 2024.
- [2] Gábor Fejes Tóth. Thinnest covering of a circle by eight, nine, or ten congruent circles. *Combinatorial and Computational Geometry*, 52:361–376, 2005.
- [3] Erich Friedman. Circles covering circles. <https://erich-friedman.github.io/packing/circovcir/>. Accessed 2026-09-14.
- [4] Richard Kershner. The number of circles covering a set. *American Journal of Mathematics*, 61(3):665–671, 1939.